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Carbon fiber implantable probe

Abstract

A carbon fiber implantable probe, a method of manufacturing the carbon implantable probe, and a method of implanting the probe in an implantation site, such as a nerve. The carbon fiber implantable probe includes a flexible probe body, a carbon fiber microarray (CFMA) composing one or more carbon fiber electrodes at least partially embedded in the flexible probe body, and a signal conductor connected to the one or more carbon fiber electrodes of the CFMA. In one example, the CFMA includes carbon fiber electrodes having conductive carbon cores partially surrounded by an insulative coating. The combination of the CFMA with the flexible probe body, made of silicone rubber for example, can improve implantation processes.

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Background/Summary

TECHNICAL FIELD

(1) This invention relates to carbon fiber implantable probes, and more particularly, to carbon fiber implantable probes having a flexible body.

BACKGROUND

(2) Long-lasting, minimally damaging, implantable probes may be used in a variety of biomedical applications, such as neural probes that interface with the peripheral nervous system. Applications for these probes range from control of robotic prostheses in cases of amputation to stimulation of autonomic nerves to regulate bladder control in cases of spinal cord injury. In addition to their stimulation capabilities, the probes may be used to record neural activity for neuroscience research or to inform the timing of applied stimulation.

(3) Most current probes include silicon shanks that penetrate the nerve and have the potential to cause considerable damage, or cuff electrodes that wrap around the outside, potentially causing less damage but also recording less discrete nerve signals. Recent work in the field has shown that recording inside the nerve produces the best signal, and that smaller electrodes produce less damage.

SUMMARY

(4) In accordance with one embodiment, there is provided a carbon fiber implantable probe, comprising: a flexible probe body; a carbon fiber microarray comprising one or more carbon fiber electrodes at least partially embedded in the flexible probe body; and a signal conductor connected to the one or more carbon fiber electrodes of the carbon fiber microarray.

(5) In accordance with various embodiments, the carbon fiber implantable probe may have any one or more of the following features, either singly or in any technically feasible combination: the flexible probe body consists of a silicone-based material; the silicone-based material is a degassed silicone rubber; the signal conductor is at least partially embedded in the flexible probe body; the signal conductor is a flex array; each carbon fiber electrode of the one or more carbon fiber electrodes has an insulative coating; the insulative coating is a functionalized polymer coating; each carbon fiber electrode of the carbon fiber microarray has an implantation end and an attachment end, wherein the attachment end is fully embedded in the flexible probe body; the implantation end, the attachment end, or both the implantation end and the attachment end include an exposed portion where a conductive carbon core is exposed; the flexible probe body includes a plurality of discrete electrode channels for each carbon fiber electrode of the carbon fiber microarray; each discrete electrode channel of the plurality of discrete electrode channels joins a conductor space; the carbon fiber microarray is configured to be deflected so that an angle θ is produced at a body interface site, with the angle θ being located between an insertion portion of each carbon fiber electrode of the carbon fiber microarray and a plane defined by an insertion end of the flexible probe body; angle θ is configured to deflect between 90° and 0° , inclusive; and/or the angle θ is configured to deflect between 90° and 45° , inclusive.

(6) In accordance with another embodiment, there is provided a method of manufacturing a carbon fiber implantable probe, comprising the steps of: aligning one or more carbon fiber electrodes in a carbon fiber microarray template; and partially embedding the one or more carbon fiber electrodes in a flexible probe body.

(7) In accordance with various embodiments, the method may have any one or more of the following steps or features, either singly or in any technically feasible combination: degassing a silicone-based material that is used for the flexible probe body; and/or the degassing step is performed before the partially embedding step.

(8) In accordance with another embodiment, there is provided a method of implanting a carbon fiber implantable probe, comprising the steps of: placing an implantation base onto a hook; using the hook to isolate an implantation site; implanting one or more carbon fiber electrodes of the carbon fiber implantable probe into the implantation site; and sealing the carbon fiber implantable

probe at the implantation site.

(9) In accordance with various embodiments, the method may have any one or more of the following steps or features, either singly or in any technically feasible combination: the hook is a nerve hook comprising: a shank portion and a bend portion, wherein the bend portion includes an implantation base cavity with two nerve cusps on opposite sides of the implantation base cavity; and/or a height of the two nerve cusps of the nerve hook is elevated with respect to the implantation base cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Preferred exemplary embodiments will hereinafter be described in conjunction with the appended drawings, wherein like designations denote like elements, and wherein:
- (2) FIG. 1 is a schematic drawing of an embodiment of a carbon fiber implantable probe;
- (3) FIGS. 2A and 2B show an embodiment of a carbon fiber microarray (CFMA) template that may be used in the manufacturing method of a carbon fiber implantable probe, such as the probe of FIG. 1;
- (4) FIGS. 3A-3C illustrate the capability of the carbon fiber implantable probe to withstand extreme lateral motion, with FIG. 3A showing the carbon fiber microarray (CFMA) before being exposed to a laterally traversing force, FIG. 3B showing the CFMA while exposed to the laterally traversing force, and FIG. 3C showing the CFMA after exposure to the laterally traversing force;
- (5) FIGS. 4A and 4B illustrate an embodiment of a carbon fiber implantable probe, with FIG. 4A showing a CFMA before forced breakage and FIG. 4B showing the CFMA after forced breakage;
- (6) FIGS. 5A and 5B illustrate another embodiment of a carbon fiber implantable probe, with FIG. 5A showing a CFMA before forced breakage and FIG. 5B showing the CFMA after forced breakage;
- (7) FIG. 6 is a flowchart illustrating an example embodiment of a method of implanting a carbon fiber implantable probe, such as the carbon fiber implantable probe of FIG. 1;
- (8) FIG. 7 shows a nerve hook that may be used to isolate the nerve for implantation;
- (9) FIG. 8 is an image of a CFMA;
- (10) FIG. 9 is an image of surgical placement using the CFMA of FIG. 8 that may be used with the implanting method of FIG. 6;
- (11) FIG. 10 shows an embodiment of a carbon fiber implantable probe as it is being inserted into a nerve; and
- (12) FIG. 11 shows an example embodiment of an implanted carbon fiber implantable probe.

DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

(13) As described herein, a carbon fiber implantable probe comprises a number of carbon fiber electrodes in a carbon fiber microarray (CFMA), with the carbon fiber electrodes being at least partially embedded in a flexible body. The combination of the CFMA and the flexible body provides an unexpected increase in the resiliency and durability of the probe, as the flexible body was shown to increase the flexibility of the carbon fiber electrodes to the point where the carbon fiber electrodes were able to withstand a 90° bend at the body interface. Moreover, implantation was possible even with extreme lateral deflection of the carbon fiber electrodes. This resiliency and durability can make the implantation process easier, and it can also help maintain a better connection between the probe and the implantation site while the probe is in use. Particular implantation techniques are also described herein, which allow for the carbon fiber implantable probe to be inserted into nerves of varying diameters with minimal handling damage to the nerve during surgery. While the description below is at least generally within the context of a peripheral nervous system application, other applications are certainly possible, such as brain-based

applications, spinal applications, muscle applications, or other organ-based applications.

(14) FIG. 1 is a schematic illustration of a carbon fiber implantable probe 20. The carbon fiber implantable probe 20 includes a plurality of carbon fiber electrodes 22 which together comprise a carbon fiber microarray (CFMA) 24. The carbon fiber electrodes 22 of the CFMA 24 are partially embedded in a flexible body 26. A signal conductor 28, such as a PCB, ribbon cable, or flex array with wires 30 (although it is certainly possible to have a wireless implementation with a WiFi transmitter or the like used as the signal conductor 28), can transmit readings to an output device 32, such as a computer or implanted device. The signal conductor 28 may also have the capability to send a stimulatory signal to one or more carbon fiber electrodes 22 of the CFMA 24. While the signal conductor 28 is shown as being embedded in the flexible body 26, other arrangements are certainly possible, such as partially-embedded or non-embedded arrangements. It should be understood that FIG. 1 is not to scale, and the relative sizes of the various components of the probe 20 are likely to vary from the illustrated embodiment. For example, the signal conductor 28 is likely to be much smaller in size as compared with the size of the flexible body 26. Similarly, the carbon fiber electrodes 22 of the CFMA 24 are likely to be much smaller in terms of length, width, etc.

(15) FIG. 1 includes an enlarged view of one of the carbon fiber electrodes 22. The carbon fiber electrode 22 includes an implantation end 34 and an attachment end 36. The implantation end 34 may have various shapes to help encourage implantation (e.g., needle-like) or retention of the device at the implantation site. The implantation end 34 and/or the attachment end 36 may include an exposed portion 38, 40 where a conductive carbon core 42 is generally exposed. The exposed portion 38, 40 is generally free of a thin, functionalized polymer coating 44 that may be used to insulate one or more portions of the conductive carbon core 42. The exposed portion 38 at the implantation end 34 may facilitate signal transmission for neuromodulation. The exposed portion 38 at the attachment end 36 may also facilitate signal transmission via connection to the signal conductor 28, such as a flex array. This connection, in some embodiments, may be accomplished with a conductive silver epoxy or a metal solder with a low melting temperature, such as indium, to cite a few examples. In other embodiments, the signal conductor 28 may be more integrally formed, patterned, layered, or the like with the conductive carbon core 42. It is possible as well to have additional exposed portions 38, 40, or to locate exposed portions at different locations along the carbon fiber electrode 22. For example, multiple exposed portions may be situated along the exposed length of the carbon fiber electrode 22.

(16) The carbon fiber electrode 22 includes an embedded portion 46 that is generally surrounded by the flexible body 26 and an insertion portion 48 that extends beyond the flexible body 26. In some embodiments, the insertion portion 48 is about 25-500 μm , preferably 100-250 μm . In some embodiments, the length of the insertion portion 48 varies between the different carbon fiber electrodes 22 in the CFMA 24 in order to accommodate insertion into nerves of different diameters or to access different depths or fascicles within a nerve. The carbon fiber electrode 22 may have a diameter at the embedded portion 46, the insertion portion 48, or both, of less than 9 μm , and in some embodiments, is about 5 μm . In some embodiments, the carbon fiber electrode 22 has an aspect ratio (defined as a length of the longest axis divided by the diameter), which is preferably about 100 and in certain implementations, greater than about 1,000, or in some embodiments, greater than 10,000. A body interface site 50 is situated between the embedded portion 46 and the insertion portion 48 where the carbon fiber electrode 22 exits the flexible body 26. As will be detailed further below, the flexible body 26 and carbon fiber electrode 22 combination can result in angles at the body interface site 50 between the embedded portion 46 and the insertion portion 48 of up to 90°.

(17) The conductive carbon core 42 may have the form of an elongated wire that spans the length of the carbon fiber electrode 22 from the implantation end 34 to the attachment end 36. In an advantageous embodiment, the conductive carbon core 42 comprises one or more carbon fibers. In

a particular implementation, the conductive carbon core **42** consists of Cytec Thorne™ T-650/35 3K carbon fiber, polyacrylonitrile (PAN) precursor. A resin matrix composite is used to treat the fiber surface, which can increase the interlaminar shear strength. In this embodiment, the conductive carbon core **42** has a 1.75% elongation at break, a modulus of elasticity of about 241 GPa, an electrical resistivity of about 0.00149 ohm-cm, and athermal conductivity of 14.0 W/m-K. In other embodiments, the conductive carbon core **42** may be comprised of a carbon fiber having 1-5% resin (or more particularly about 2% resin), glassy carbon microstructures, carbon nanotubes (CNTs), metallic CNTs, or a CNT composite. Certain desirable carbon fibers can have a modulus of elasticity of greater than or equal to about 200 GPa, for example between 240 GPa to about 999 GPa.

(18) The coating **44** on the carbon fiber electrodes **22** may be an insulative coating; however, in some embodiments, a coating may not be used at all, and one or more of the carbon fiber electrodes **22** may be uncoated. An uncoated carbon fiber electrode **22** may be used in implementations where the flexible body **26** is sufficiently insulative. The coating **44** is advantageously deposited, such as via a chemical vapor deposition (CVD) process, a micro-patterning process, or another suitable coating process. In one embodiment, the coating consists of parylene, or more particularly, parylene-c. In another embodiment, the coating includes functionalized poly-p-xylylenes. Poly(3,4-ethylenedioxythiophene):p-toluene sulfonate (PEDOT:pTS) may be selectively coated onto one or more areas of the carbon fiber electrode **22**, such as at the exposed portion **38** and/or the implantation end **34**. In some embodiments, only selected areas are coated, such as embodiments having multiple exposed portions along the insertion portion **48**. For example, platinum black may be selectively electrodeposited as recording sites on the insertion portion **48** of the conductive carbon core **42**. In one embodiment, the conductive carbon core **42** is nominally about 5 μm on a side or in diameter, and has a coating layer that is about 0.5-1 μm thick. Other coating materials, processes, and/or coating configurations are certainly possible.

(19) A plurality of carbon fiber electrodes **22** form the carbon fiber microarray (CFMA) **24**. As shown in FIG. 1, the CFMA **24** may include a relatively straight line of carbon fiber electrodes **22** that are evenly spaced or distributed in the flexible body **26**. This configuration may be advantageous for minimally invasive methods of PNS neuromodulation. In other embodiments, the CFMA **24** may have a different configuration. For example, the carbon fiber electrodes **22** may be clustered in brain or organ-based implementations. However, the line configuration is desirable in embodiments where the probe **20** is used to obtain readings from a single nerve or neuron. In some embodiments, carbon fiber electrodes **22** in a CFMA **24** may have lateral offsets (1-200 μm between electrodes) to interface with different clusters of neurons across the span or width of a nerve. Further, the CFMA **24** may have more or less carbon fiber electrodes **22** than what is schematically illustrated in FIG. 1.

(20) The flexible body **26** works unexpectedly well with the CFMA **24**. As opposed to typical or standard silicon shanks or probes, which are comparatively much more rigid, the flexible body **26** increases the flexibility of the carbon fiber electrodes **22** to the point where the electrodes can withstand a 90° bend at the body interface site **50**. “Flexible,” when used to describe the body **26**, means that the body is comprised of a rubber or thermoset elastomer (TSE) in one embodiment. In a preferred embodiment, the flexible body **26** is silicone, or even more preferably, A-103 Medical Grade Elastomer from Factor II, Inc. This particular silicone is a two-component product, which, when combined, cures to a translucent silicone rubber. The elastomer component consists of a dimethylsiloxane polymer, a reinforcing silica, and a platinum catalyst. The curing agent component consists of a dimethyl-siloxane polymer, an inhibitor, and a siloxane cross linker. Other materials are certainly possible, such as parylene, polyamide, polydimethylsiloxane (PDMS), or SU-8, to cite a few examples.

(21) In another embodiment, “flexible,” when used to describe the body **26**, means that the body has a modulus of elasticity between 0.000005-23 GPa. In a preferred embodiment, the body **26** has

a modulus of elasticity between about 0.07 and 3 GPa. This modulus of elasticity results in a much more flexible body than comparable silicon shanks, which have a modulus of elasticity of about 60 GPa or more. In yet another embodiment, “flexible,” when used to describe the body **26**, means that the body has an elongation at break of about 60-1120%. In a preferred embodiment, the body **26** has an elongation at break of about 700-900%. Again, compared with typical silicon probes, which have an elongation at break that is typically less than 10%, the flexible body **26** can more resiliently support the CFMA **24**.

(22) The flexible body **26** includes an insertion end **52** and a distal end **54** with an outer side surface **56** therebetween. Other shapes are certainly possible. For example, the probe **20** could include a number of sidewalls instead of the continuous outer side surface **56**. To accommodate the CFMA **24**, the flexible body **26** includes a plurality of discrete electrode channels **58**, each of which accommodates one of the carbon fiber electrodes **22**. To accommodate the signal conductor **28**, the flexible body **26** includes a conductor space **60**. Each electrode channel **58** extends from the insertion end **52** to the conductor space **60**. The electrode channels **58** are generally cylindrically shaped, but this may or may not vary depending on the shape of the carbon fiber electrode **22**. The flexible body **26** can provide for a more stress-resistant electrode channel **58**, allowing for relative movement between the CFMA **24** and the insertion end **52** at each body interface site **50**.

(23) To manufacture the carbon fiber implantable probe **20**, a CFMA template **62**, one example of which is shown in FIG. 2A, may be used. The CFMA template **62** may align the carbon fiber electrodes **22** in the CFMA **24** while they are embedded in the flexible body **26**. The CFMA template **62** is a zero insertion force (ZIF) board, which is positioned over a mold **63**. An enlarged view of the mold **63** is shown in FIG. 2B. The carbon fiber electrodes **22** are mounted to the ZIF board for alignment purposes with silver epoxy and extend to a length of about 1.5 mm. The mold **63** for the flexible body **26** is an aluminum sheet with a plurality drilled depressions for a silicone flexible body, is filled with the body material to embed the CFMA **24**. The mold **63** can be coated with a thin layer of lubrication jelly to facilitate removal. In one embodiment, detailed below, the uncured or partially cured flexible body material is degassed. The CFMA template **62** or ZIF board is lowered into the mold until confirmation of slight fiber bending, indicating that the carbon fiber electrodes **22** have reached the depth of the mold. The uncured or partially cured material of the flexible body **26** is then cured. In one embodiment, where the flexible body **26** comprises silicone, the body material is cured for about 10-12 minutes at 115° C. Microforceps or another cutting mechanism can be run across the carbon fiber electrodes **22** at the edge of the ZIF board with applied pressure to break the carbon fiber electrodes **22** at the implantation end **34**. In another embodiment, the carbon fiber electrodes are cut again to define the implantation end **34**.

Alternatively, the CFMA template **62** may be an SU-8 jig or some other apparatus to properly position the carbon fiber electrodes **22**. In yet other embodiments, the CFMA template **62**, the probe **20**, and/or the flexible body **26** is 3D printed. Other manufacturing processes are certainly possible.

(24) In some embodiments, the flexible body **26** comprises a degassed silicone. When the silicone is heated to cure, air bubbles in the silicone can expand, and if these air bubbles are in the vicinity of the carbon fiber electrodes **22**, the carbon fiber electrodes **22** may move from their originally straight position. Degassing the silicone allows for the formation of a straight electrode channel **58** to accommodate a straight carbon fiber electrode **22**. Further, the degassed silicone has a higher density which can provide more structural support for the CFMA **24**. In one embodiment, during the manufacturing process, filled molds are degassed in a vacuum chamber for approximately 40-60 minutes with periodic vacuum release until the visible bubbles are gone. The CFMA template **62**, such as the ZIF board described above, can then be aligned above the mold and then lowered into the uncured material of the flexible body **26** and held in the body until the material is cured. Degassing can be accomplished before or after embedding. Before embedding may be preferred in some instances, as the degassing process may cause the carbon fiber electrodes **22** to be shifted

from their straight position orthogonal to the insertion end **52** of the flexible body **26**.

(25) FIGS. **3A-3C** show how the flexible body **26** can increase the durability of the carbon fiber electrodes **22** (only one is labeled in each of FIGS. **3A-3C** for clarity purposes), thereby increasing the surgical implementation capability of the CFMA **24**. In this embodiment, the insertion portion **48** of each of the carbon fiber electrodes **22** is 245 μm . A glass capillary **64** was laterally moved across the insertion end **52** of the flexible body **26**. The size of the capillary **64** was chosen to approximate the size of a large nerve. FIG. **3A** shows the CFMA **24** before contact with the capillary **64**. FIG. **3B** shows the CFMA **24** as its being deflected by the capillary **64**. As shown, the CFMA **24** is deflected such that an angle θ is produced at the body interface site **50**, with the angle θ being located between the insertion portion **48** of each carbon fiber electrode **22** and a plane defined by the insertion end **52** of the body **26**. The angle θ was able to change from 0° , to about 45° as shown in FIG. **3B**, to 90° , and then back to 0° as shown in FIG. **3C**. The CFMA **24** was unharmed after a 90° deflection.

(26) FIGS. **4A-4B** and **5A-5B** also show how the flexible body **26** can increase the durability of the carbon fiber electrodes **22**. In FIGS. **4A** and **4B**, the CFMA **24** is approximately 125 μm long. In FIGS. **5A** and **5B**, the CFMA **24** is approximately 200 μm long. FIGS. **4A** and **5A** show each CFMA **24**, respectively, before a forced breakage of a plurality of the carbon fiber electrodes **22**. Each CFMA **24** was then exposed to a breaking force, which caused the carbon fiber electrodes **22** to be oriented at various angles θ relative to the insertion end **52**, or in some cases, caused the carbon fiber electrodes **22** to be broken off altogether. FIGS. **4B** and **5B** show each probe **20** after forced breakage. Unexpectedly, both probes in FIGS. **4B** and **5B**, with the various angles θ , were successfully implanted.

(27) FIG. **6** is a flowchart illustrating example steps of a method **100** of implanting a carbon fiber implantable probe, such as the carbon fiber implantable probe **20** schematically illustrated in FIG. **1**. Step **102** involves placing an implantation base onto a hook, which is an optional step. An example hook is the nerve hook **70** illustrated in FIG. **7**. The nerve hook **70** includes a shank portion **72** and a bend portion **74**. The bend portion **74** ends at a tip **76** such that a gap **78** exists between the tip **76** and the shank portion **72**. The length of the gap **78** may vary and can depend on the size of the target nerve for implantation. The size of gap **78** also allows for ample space from above for the probe **20** to be inserted. The bend portion **74** includes an implantation base cavity **80** for placing the implantation base, such as a silicone piece, that is used later in the method **100** for implanting the carbon fiber implantable probe **20**. Two opposing walls of the implantation base cavity **80** include nerve cusps **82**, **84**, which can help position an isolated nerve. Alternatively, the base cavity **80** may have a deeper recession, or well, that allows drop casted material to flow around the nerve and probe **20**. The height of the nerve cusps **82**, **84** may be elevated relative to the base cavity **80** to provide greater isolation of the nerve to potentially minimize water interference on the nerve surface during insertion. This height may be adapted in various nerve hook embodiments to account for dimensions of the implantation site or other surgical considerations.

(28) Step **104** of the implantation method **100** involves using the hook **70** to isolate an implantation site. FIG. **8** shows the probe **20**, and FIG. **9** shows surgical placement of the non-functional probe **20** of FIG. **8** into a rat cervical vagus nerve. The nerve hook **70** is lowered toward the implantation site and used to lift the vagus nerve from the surrounding tissue. FIG. **9** is an enlarged view of the implantation site **88**, as hooked by the hook **70** with an optional implantation base **90**. FIG. **10** shows another embodiment of a hook. In this embodiment, a dual-pronged glass hook **92** is used to isolate the implantation site. Since it may be difficult to hold an implantation base with the glass hook **92**, the nerve hook **70** may be preferred in surgeries where an implantation base is not necessary or desired. Other implantation assist devices are certainly capable of being used.

(29) Returning to FIG. **6**, step **106** of the implantation method **100** involves implanting one or more carbon fiber electrodes **22** of the CFMA **24** of the probe **20** into the implantation site, such as the implantation site **88**. The probe **20** can be carried by a suction wand **94** at the distal end of the body

and lowered toward the hook **70**, as shown in FIG. **9**. A small blunt needle tip attached to the suction wand **94** may be used to pick up the probe **20**. As described above, the flexible body **26** allows for greater implantation and surgical freedom, as bending or breaking of one or more of the carbon fiber electrodes **22** can be tolerated without undesirably impacting the recording and/or stimulating capabilities of the probe **20**.

(30) Step **108** of the method involves sealing the probe **20** to the nerve at the implantation site **88** using a drop cast sealant. FIG. **11** shows a casted implantation site **96** where a biocompatible liquid silicone has been applied to the nerve hook **70**, covering both the flexible body **26** of the probe **20**, as well as the implantation base **90**, thereby sealing the probe **20** and implantation base **90**. In one embodiment, Kwik-Sil™ is used to seal the probe **20** to the nerve. In another embodiment, fibrin glue is used to seal the probe **20** to the nerve. If a needle tip is used to initially help pick up the probe **20**, the needle tip may be removed from the probe during the curing process. After curing, the implanted nerve and casted implantation site **96** are lifted off the nerve hook **70** and placed back in the cavity. Other steps are certainly possible, such as using a removable PEG layer on an insertion rod to insert the probe **20**, tying off pre-sewn sutures for additional stability after drop casting, or wrapping the cast implantation site and nerve with a porcine wrap, to cite a few examples.

(31) It is to be understood that the foregoing description is of one or more preferred exemplary embodiments of the invention. The invention is not limited to the particular embodiment(s) disclosed herein, but rather is defined solely by the claims below. Furthermore, the statements contained in the foregoing description relate to particular embodiments and are not to be construed as limitations on the scope of the invention or on the definition of terms used in the claims, except where a term or phrase is expressly defined above. Various other embodiments and various changes and modifications to the disclosed embodiment(s) will become apparent to those skilled in the art. All such other embodiments, changes, and modifications are intended to come within the scope of the appended claims.

(32) As used in this specification and claims, the terms “for example,” “e.g.,” “for instance,” and “such as,” and the verbs “comprising,” “having,” “including,” and their other verb forms, when used in conjunction with a listing of one or more components or other items, are each to be construed as open-ended, meaning that the listing is not to be considered as excluding other, additional components or items. Other terms are to be construed using their broadest reasonable meaning unless they are used in a context that requires a different interpretation.

Claims

1. A carbon fiber implantable probe, comprising: a flexible probe body; a carbon fiber microarray comprising a plurality of carbon fiber electrodes at least partially embedded in the flexible probe body, wherein each carbon fiber electrode of the carbon fiber microarray has an implantation end and an attachment end, wherein the attachment end is fully embedded in the flexible probe body and the implantation end is fully exposed and separated from the flexible body, wherein, for each of at least some of the carbon fiber electrodes, the implantation end and attachment end are spaced by a distance equal to a longest length of the carbon fiber electrode, wherein a body interface site is formed where each carbon fiber electrode of the carbon fiber microarray exits the flexible probe body, and wherein the flexible probe body allows for relative movement between the carbon fiber microarray and an insertion end of the flexible probe body at each body interface site, wherein one or more carbon fiber electrodes of the plurality of carbon fiber electrodes are moveable with respect one or more other carbon fiber electrodes of the plurality of carbon fiber electrodes, wherein the carbon fiber microarray is configured to implant one or more of the carbon fiber electrodes between the insertion end and the flexible probe body into an implantation site, and wherein the flexible probe body is flexible at the insertion end where each carbon fiber electrode of

- the carbon fiber microarray exits the flexible probe body, wherein flexible is a material having a modulus of elasticity between 0.000005 GPa and 23 GPa; and a signal conductor connected to the one or more carbon fiber electrodes of the carbon fiber microarray.
2. The carbon fiber implantable probe of claim 1, wherein the flexible probe body consists of a silicone-based material.
 3. The carbon fiber implantable probe of claim 2, wherein the silicone-based material is a degassed silicone rubber.
 4. The carbon fiber implantable probe of claim 1, wherein the signal conductor is at least partially embedded in the flexible probe body.
 5. The carbon fiber implantable probe of claim 4, wherein the signal conductor is a flex array.
 6. The carbon fiber implantable probe of claim 1, wherein each carbon fiber electrode of the one or more carbon fiber electrodes has an insulative coating.
 7. The carbon fiber implantable probe of claim 6, wherein the insulative coating is a functionalized polymer coating.
 8. The carbon fiber implantable probe of claim 1, wherein the implantation end, the attachment end, or both the implantation end and the attachment end include an exposed portion where a conductive carbon core is exposed.
 9. The carbon fiber implantable probe of claim 1, wherein the flexible probe body includes a plurality of discrete electrode channels for each carbon fiber electrode of the carbon fiber microarray.
 10. The carbon fiber implantable probe of claim 9, wherein each discrete electrode channel of the plurality of discrete electrode channels joins a conductor space.
 11. The carbon fiber implantable probe of claim 1, wherein the carbon fiber microarray is configured to be deflected so that an angle θ is produced at the body interface site, with the angle θ being located between an insertion portion of each carbon fiber electrode of the carbon fiber microarray and a plane defined by the insertion end of the flexible probe body.
 12. The carbon fiber implantable probe of claim 11, wherein the angle θ is configured to deflect between 90° and 0° , inclusive.
 13. The carbon fiber implantable probe of claim 12, wherein the angle θ is configured to deflect between 90° and 45° , inclusive.
 14. A method of implanting the carbon fiber implantable probe of claim 1, comprising the steps of: placing an implantation base onto a hook, wherein the implantation base is separate from the carbon fiber implantable probe; using the hook to isolate the implantation site; implanting one or more carbon fiber electrodes of the carbon fiber implantable probe into the implantation site; and sealing the carbon fiber implantable probe at the implantation site.
 15. The method of claim 14, wherein the hook is a nerve hook comprising: a shank portion; and a bend portion, wherein the bend portion includes an implantation base cavity with two nerve cusps on opposite sides of the implantation base cavity, wherein the implantation base is placed into the implantation base cavity in the bend portion of the nerve hook.
 16. The method of claim 15, wherein a height of the two nerve cusps of the nerve hook is elevated with respect to the implantation base cavity.
 17. The carbon fiber implantable probe of claim 1, wherein the plurality of carbon fiber electrodes includes eight or more carbon fiber electrodes.
 18. The carbon fiber implantable probe of claim 1, wherein each carbon fiber electrode of the plurality of carbon fiber electrodes has an aspect ratio which is greater than 1,000.
 19. The carbon fiber implantable probe of claim 18, wherein each carbon fiber electrode of the plurality of carbon fiber electrodes has an aspect ratio which is greater than 10,000.
 20. The carbon fiber implantable probe of claim 1, wherein a length of the insertion portion varies between one or more carbon fiber electrodes of the plurality of carbon fiber electrodes.
 21. A method of manufacturing a carbon fiber implantable probe, comprising the steps of: aligning

a plurality of carbon fiber electrodes in a carbon fiber microarray template; degassing a silicone-based material that is used for a flexible probe body; and partially embedding the plurality of carbon fiber electrodes in the flexible probe body after aligning the plurality of carbon fiber electrodes in the carbon fiber microarray template to form a carbon fiber microarray that is at least partially embedded in the flexible probe body, wherein a body interface site is formed where each carbon fiber electrode of the carbon fiber microarray exits the flexible probe body, and wherein the flexible probe body allows for relative movement between the carbon fiber microarray and an insertion end of the flexible probe body at each body interface site, wherein one or more carbon fiber electrodes of the plurality of carbon fiber electrodes are moveable with respect one or more other carbon fiber electrodes of the plurality of carbon fiber electrodes, wherein the carbon fiber microarray is configured to implant one or more of the carbon fiber electrodes between the insertion end and the flexible probe body into an implantation site, and wherein the flexible probe body is flexible at the insertion end where each carbon fiber electrode of the carbon fiber microarray exits the flexible probe body, wherein flexible is a material having a modulus of elasticity between 0.000005 GPa and 23 GPa.

22. The method of claim 21, wherein the degassing step is performed before the partially embedding step.
